

High Potential: GenC Accelerated Learning Program - Student Handbook

Fast-Track Your Career. Unlock Your Potential.

Mode: Fast-Track | Business-Ready | Project-Aligned

HiPo - Quality Engineer with GenAI based Selenium Automation - v1



Program Overview

Welcome to the High Potential (HiPo) GenC Accelerated Advanced Learning Program — a fast-paced and structured initiative designed to identify and nurture high-performing learners early in their training journey.

The program focuses on recognizing GenCs who exhibit strong technical and behavioral potential, enabling them to fast-track through the learning path, complete training earlier, and be ready for business deployment ahead of schedule.

This handbook will guide you through the program's structure, evaluation process, and key expectations from you as a learner.

Program Objectives

The **HiPo GenC Accelerated Learning Program** aims to:

- Provided opportunity to learners with strong foundational and behavioural skills aligned to the curriculum.
- Enable eligible learners to move into advanced learning stages through a **Fast-Track Learning Path**.
- Early deployment to niche projects
- Reduce skilling cycle time and accelerate business readiness

Benefits

The program offers measurable benefits for both learners and the business:

- **Early deployment** of HiPo GenCs for niche or critical project requirements.
- **Reduced training cycle** by 2 to 4 weeks for qualified learners.
- **Merit-based advancement**, ensuring faster progression through demonstrated capability.

What Is It?

The Boot Camp is your official welcome to the organization and the HiPo journey. It's not just about learning - it's about immersing yourself in the culture, expectations, and mindset of a future-ready engineer.

What You would have experienced

- **Organizational Learning Sessions**
Understand the company's mission, values, and how your role fits into the bigger picture.
- **Behavioral Training Workshops**
Led by certified trainers and coaches, these sessions focus on communication, collaboration, adaptability, and ownership.
- **Leader Connect Sessions**
Interact with business leaders and delivery managers to gain insights into real-world projects and expectations.
- **Stakeholder Mapping**
Learn about key stakeholders - HR, L&D, BU leads, mentors—and how to engage with them effectively.
- **Training Program Orientation**
Get a walkthrough of your fast-track curriculum, milestones, and evaluation checkpoints.
- **Policy Awareness**
Understand essential policies around ethics, conduct, data security, and workplace behavior.

Behavioral Assessment

At the end of the boot camp, you have undergone a structured behavioural assessment to evaluate:

- Critical thinking and decision-making
- Problem-solving approach
- Team collaboration and communication
- Professionalism and accountability

Why It Matters

This phase ensures that HiPo GenCs are not just technically sound, but also culturally aligned, behaviorally mature, and business ready. It sets the tone for your learning journey and helps identify those who are ready to lead from the front.

Happy to see you have accomplished below Checkpoint Zero: Foundation-Ready Program

Before you enter the fast lane of the HiPo GenC Accelerated Learning Program, you have faced a **technical assessment** designed to validate your readiness in core foundational skills. This is your moment to prove that you've built a solid base—and are ready to scale up!

What's Being Evaluated?

You had been tested on the essential building blocks of QA Automation Engineer:

- **Core Java (v21)** – OOPs, collections, exception handling, streams, and functional programming,
- **Data Structures and Algorithms** –
 - Arrays, Array Lists, Linked Lists, Doubly Linked Lists, Stacks & Queues, Binary Trees, Hash Tables, Heaps, Graphs)
 - Dynamic Programming, Sorting, Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort
 - Searching: Breadth First Search, Depth First Search
- **HTML5, CSS3, JavaScript** – Web structure, styling, and interactivity.
- **ANSI SQL with MySQL** – Data modeling, queries, joins, and constraints.
- **Maven Fundamentals**
- **Functional Testing** – Manual Testing, STLC, Agile Process, Jira
- **Selenium WebDriver, TestNG**

Why It Matters

This assessment helps us identify GenCs who are ready to:

- Skip the basics and jump straight into advanced engineering.
- Handle real-world project scenarios with confidence.
- Learn faster, build smarter, and deploy sooner.

Your Goal

Cleared the foundational assessment and evaluation with confidence and earned your spot in the HiPo fast-track batch. This is your launchpad to advanced learning, early graduation, and deployment.

Batch Formation & Fast-Track Pathway

Based on performance in the evaluation:

- Qualified learners will be grouped into a **HiPo Batch** and transitioned to the **Fast-Track Learning Path**, featuring an accelerated progression schedule.
- Others will continue in the **Standard Learning Path**, receiving additional mentoring and support to strengthen their skills.

Both learning paths lead to the same final goal — achieving QA Automation Developer readiness. The difference lies in pace and independence.

Learning Path Mapping & Final Evaluation

HiPo GenCs in the **Fast-Track Path** will continue an accelerated schedule, covering advanced concepts earlier and engaging in deeper application exercises.

At the end of the training journey, all learners will undertake a **Final Evaluation** designed to validate:

- Mastery Java concepts and Data structures Algorithms for automation
- Capability to test automation development and troubleshoot applications
- Readiness for project deployment and business integration.

Successful completion of this evaluation marks your **graduation from the program** and transition to the business environment.

Recommended Program Sequence in the learning path

- **STAGE: Core Java With SQL_Refresher**

Congratulations on successfully completing the Foundation Program, including both the behavioural and technical assessments, as well as the interview process! We are excited to onboard you to the advanced learning program: STAGE: Functional Testing with QA GenAI-based Selenium Automation. To help you succeed in this advanced program, a strong foundation in programming and databases is essential. Therefore, we are providing access to an optional refresher module STAGE: Core Java with SQL Refresher. This module is available for you to revisit and strengthen your skills at any time, based on your individual learning needs.

- **STAGE: Functional testing With QA GenAI based Selenium Automation**

Welcome to the agile automation journey! You would master core Selenium and functional testing within an agile context, including templates like user stories and sprint backlogs. The next phase integrates advanced tools like TestNG, POM, Docker, BDD, Devops and CI/CD

for comprehensive, API Automation and real-world framework development. We will apply all learned modules to automate the functional requirements. This culminates in building a robust, client-ready hybrid automation framework from scratch. It is mandatory for you to accomplish successfully this stage with BU SME interview

- **STAGE: GenAI Code Assist (GitHub Copilot)**

Learning and practicing the below provided GenAI topics will equip you with essential, modern skills in QA Automation, preparing you for an evolving tech landscape

- **Prompt Engineering & GitHub Copilot(GenAI Foundation):** This section provides the foundational knowledge of how GenAI works and the critical importance of using it ethically and responsibly.
- **Prompt Engineering:** The ability to communicate effectively with AI models is a core competency for maximizing their utility. This skill directly translates to generating higher quality code suggestions, more relevant test cases, and accurate documentation, saving significant time and effort.
- **GitHub Copilot:** This section focuses on integrating AI assistance directly into the daily workflow. Learning to use GitHub Copilot effectively helps freshers write code faster, reduce bugs through AI-driven suggestions, and understand best practices through examples provided by the AI.
- **Usecase and Handbook & Comprehensive GenAI Hands-On for Automation Developers:** These exercises provide practical experience in optimizing AI outputs, managing token limitations, and generating functional, high-quality automation code, making the learning directly applicable to their job roles.

Keys to Success

Area	What's Expected
Technical Mastery	Solid understanding of Functional testing With QA GenAI based Selenium Automation
Analytical Thinking	Logical reasoning and structured problem-solving.
Code Quality	Clean, readable, and maintainable code practices.

Learning Agility	Quick adaptability and self-learning initiative.
Professionalism	Discipline, accountability, and teamwork mindset.

Skills to Prepare

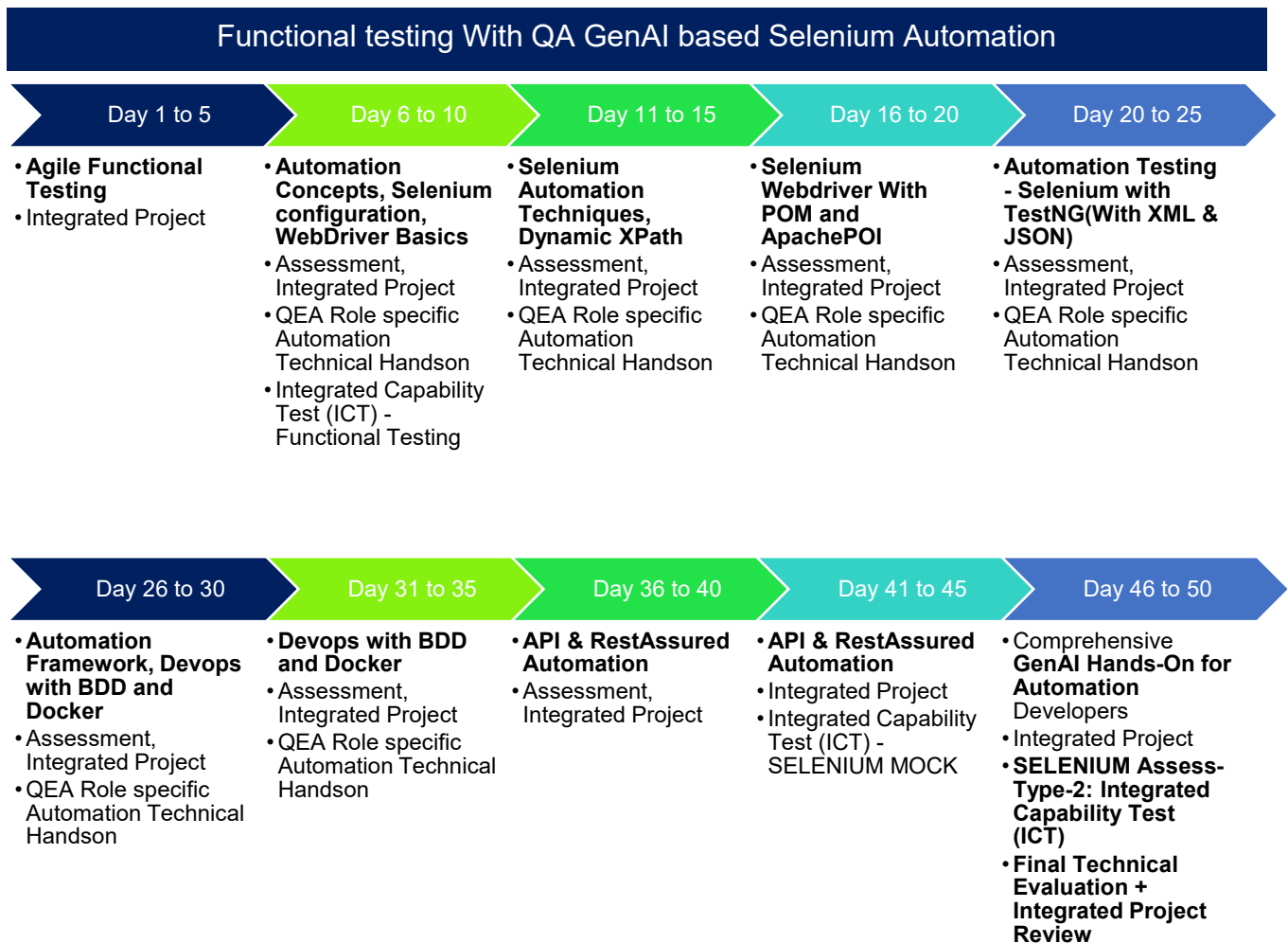
Before and during the early phase of the program, you are encouraged to strengthen your understanding of the following areas to maximize your performance in upcoming evaluations:

Skill Area	Focus Topics
Advanced QA Development technologies	Agile Functional Testing Automation Concepts, configuration, WebDriver Basics Selenium Automation Techniques, Dynamic XPath Selenium Webdriver With POM and ApachePOI Automation Testing - Selenium with TestNG Automation Framework Devops with BDD and Docker API Automation Comprehensive GenAI Hands-On for Automation Developers
Database Fundamentals	ANSI SQL using MySQL
Programming in Java	Core Java (v21), Object-Oriented Concepts, Exception Handling, Collections, Streams
Data Structures and Algorithms	Data Structures and Algorithms, Dynamic Programming, Searching
Build & Dependency Management	Maven Fundamentals
Design Principles	SOLID Principles and Clean Code Practices

Tracking and Project Management tools	JIRA
Debugging Skills	Troubleshooting and Problem Analysis

Start early — familiarize yourself with documentation, hands-on exercises, and small projects to build confidence before formal evaluations begin.

Program Highlights



Program Completion Criteria

100% Completion of Hands On in Stage 2 is mandatory for interim / final evaluation eligibility.

During the final evaluation, the GenC will be interviewed by a Technical SME from the BU to assess your knowledge through a technical discussion. **In the same evaluation call, the Integrated project completed by individual will be evaluated by the BU SME.**

You are expected to gain RAG status as Green to complete this **HiPo GenC Accelerated Learning Program** successfully

For the final evaluation, the GenC will participate in a video interview conducted by a Tech SME from the BU.

Table - Check Point Intervals

Check points	Interpreting Status
Final Evaluation	- Green - On Track for Graduation - Red /Amber – Only 2 attempts are given Note: If student fails after the applicable re-attempts, they will be considered as “Not-Graduated”.

Software Requirements

Below are the software's need to be installed in system from Company Portal, before start proceeding with selenium Handson and projects like mini project and hackathon.

- Non Catalog - Eclipse Org Eclipse IDE 2025-03 R Windows
- Selenium Java 4.27 Windows Open Source
- The Apache POI 5.3
- TestNG for Eclipse 7.9.0 Freeware
- Non Catalog - Postman, Inc. Postman for Windows 11.42.4
- MySQL Installer - Community 8.0.41 Windows
- Non Catalog - Jenkins project Jenkins 2.497
- Zulu OpenJDK 23.30.13 Window

NOTE: If genc's are facing any challenges while installation, Coach/trainer/mentor will help to install the software.

Objective to use the software:

- Developing the project
- Writing the code
- Debugging the errors
- Testing the applications

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Day 1 to 5

Agile Functional Testing

Continuous Learning: Technical Enablement

Learn the basics of Agile Fundamentals and Software Testing Life Cycle



[Learn Manual Software Testing + Agile with Jira Tool](#)

Learn all the sections in this Udemy course to gain deep understanding about below modules

Module : Software Testing Concepts

- Understanding Software Development Life Cycle (SDLC) and Software Testing Life Cycle (STLC)
- Distinguishing between Project and Product in the context of testing
- Exploring V-Model and the roles of QA (Quality Assurance), QC (Quality Control), and QE (Quality Engineering)
- Differentiating between White Box and Black Box Testing
- Delving into Static Testing and Dynamic Testing
- Conceptualizing Verification and Validation in software testing
- System Testing Types, including GUI Testing and Functional vs. Non-Functional Testing
- Test Design Techniques and strategies like Re-Testing, Regression Testing, Exploratory Testing, Adhoc Testing, Sanity, Smoke Testing, and End-To-End Testing
- Understanding Software Testing Life Cycle (STLC) phases, including Use case, Test scenario, Test case, Test Environment, Execution, Defect Reporting, Test Closure, and Test Metrics.

Module : Software Testing Project

- Introduction to project
- Extracting Functional Requirements from Functional Requirement Specifications (FRS)
- Creating comprehensive Test Scenarios and Test Cases
- Execution of Test Cases
- Reporting and Tracking Bugs
- Test Sign-off

Phase : Agile Testing + Jira Tool

Agile/Scrum Process:

- Overview of Agile and Scrum methodologies
- Understanding Scrum Teams, Sprints, and User Stories
- Estimating User Stories and defining 'Definition of Done' and 'Definition of Ready'
- Exploring Sprint Activities such as Sprint Planning, Backlog Refinement, Sprint Review, and Sprint Retrospective

Jira Tool:

- Installation and Configuration of JIRA
- Project creation in Jira and user management
- Creating Backlog, Epic, and User Stories in JIRA
- Sprint creation and management
- Understanding the Sprint life cycle in JIRA
- Bug creation in Jira
- Writing test cases in JIRA with Zephyr plugin
- Creating Test Cycles and executing/updating Test Cases in Jira

Continuous Learning: Technical Hands-on

Mandatory Hands-on

- The Importance of QEA and Career Scope
 - The Scope Debate: Quality Assurance vs. Software Development
 - Trainer-Guided Integrated Project: Activity Readiness
 - Trainer-Guided Integrated Project: Configured Jira Project & Backlog Management
 - Trainer-Guided Integrated Project: Test Case Development in Jira (using Zephyr/similar plugin)
 - Trainer-Guided Integrated Project: Test Execution & Status Reporting Artifacts
 - Trainer-Guided Integrated Project: User story document
 - Trainer-Guided Integrated Project: Retrospection Report
 - Trainer-Guided Integrated Project: Sprint Backlog
 - Placeholder to submit other Functional agile deliverables
-
- ✓ **You should have introduced more than one sprints**
 - ✓ **Ensure you have in-sprint automation concepts introduced as part of project**
 - ✓ **Ensure you have created Regression suite and smoke suite**
 - ✓ **Ensure your regression suite consists of minimum 25 test cases and test scripts**
 - ✓ **Ensure your smoke suite covers end to end business functionality**

Day 6 to 10

Day 6 to 10 will be focusing on Selenium configuration and WebDriver Basics.

Udemy learnings are recommended in the Platform to understand the fundamental concepts.

Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Integrated Project Deliverable: Agile and Functional Testing handson

As the selenium learning starts, the project details(By trainer) will be given for the learners so that you can parallel keep doing the project activities along with the rest of the learnings.

Please proceed with the automation implementation for the functional requirements provided in the previous milestone ("Trainer Given Handson").

Instructions: As you progress through this milestone's learning objectives (Selenium Architecture, Environment Setup, WebDriver, Locators, and Customized XPATH), incrementally integrate these newly acquired skills into your existing automation project.

Objective: The goal is to build a foundational, executable set of automated test scripts that validate the functionality defined in the agile documents you previously prepared.

The given Project are expected to be completed on or before the end week of the program.

Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

Automation Concepts, configuration, WebDriver Basics

Learn the basics of WebDriver Basics



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 22 to 35

Section 3: Selenium WebDriver

Selenium - Course - TOC			
Chapter	Chapter Name	Topic	Training Mode
Chapter 1	Selenium Basics, Selenium Architecture & Setting up Selenium environment	What is Selenium and Overview	In-Person Trainer - ILT (ILT - Instructor Led Training)
		Selenium Components (WebDriver vs Selenium IDE vs Selenium Grid)	
		Understanding the WebDriver Architecture	
		Installing Java Development Kit (JDK), Configuring Selenium WebDriver and Eclipse IDE	
		Locating GUI Elements in Application Under Test (AUT)	
		How to identify the locators in web page & Launching Browsers in Selenium	
		HandsON : TekStac Lab	Daily Lab Exercise(DLE)
Chapter 2	Locating various Web elements	Find Element by Id,Name,Link Text,Partial Link Text,Class Name,CSS (Tag , ID , Class , Attribute , etc..)	In-Person Trainer - ILT
		Handling Shadow DOM in Selenium	In-Person Trainer - ILT
		Relative Locators and Chained Locators in Selenium	
		Locating By Xpath (Single Slash, Double Slash , contains(), starts_with(), text(), Last())	
		HandsON : TekStac Lab	DLE & BU Connects
Chapter 3	Desired capabilities, selenium webDriver commands and Web elements interactions	Find options in dropdown. Handle dropdown without using Select Class, Find Radio Element	In-Person Trainer - ILT
		Desired Capabilities in Selenium WebDriver	
		Selenium WebDriver- Commands	
		Text fields and buttons, Find Element VS FindElements in Selenium WebDriver - check order	
		Handling Canvas Elements in Selenium	In-Person Trainer - ILT
		Using JavaScript Executor in Selenium	
		HandsON : TekStac Lab	DLE & BU Connects
Chapter 4	WebDriver Actions	Mouse and keyboard actions	In-Person Trainer - ILT
		Drag and drop	
		Handling frames and windows	
		mouse hover (tool tip) in Selenium	
		Handling tabs & cookies	
		HandsON : TekStac Lab	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Refer to and learn the same topics covered by the trainer on Udemy.
2. Practice the Tekstac Platform Hands-on.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Find Element by Id
- Find Element by Name
- Find options in dropdown
- Find Radio Element
- Find Element by Tag Name
- Handling Check Box
- Find Element by Class Name
- LinkText and PartialLinkText

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Automation Feasibility Report & Strategy
- Trainer-Guided Integrated Project: Configured Automation Environment & Project Structure
- Trainer-Guided Integrated Project: Automated Test Scripts (Individual Functional Scenarios)

- Trainer-Guided Integrated Project: Test Execution Reports and Status
- Trainer-Guided Integrated Project: Code Review and Feedback Session Artifacts

QEA Role specific Automation Technical Handson

Mandatory Hands-on

- Chapter 2_Handson
- Chapter 3_Handson
- Chapter 4_Handson
- Chapter 5_Handson

Once completing Udemy course, Learnings, all Handson & Practice assessments related to Selenium-Web Driver skill appear for the Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge (Day 9)

Assess-Type-1:Code Challenge - Automation Concepts, Selenium configuration, WebDriver Basics

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

Functional Testing ICT (Day 10)

Assessment Duration: 4 hrs

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

Integrated Capability Test (ICT) - Functional Testing

Assess Type 2 Skill Based Assessment.

Day 11 to 15

Selenium Automation Techniques, Dynamic XPath

Learn the basics of WebDriver Basics



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 36 to 40

Section 3: Selenium WebDriver

Selenium - Course - TOC			
Chapter	Chapter Name	Topic	Training Mode
Chapter 5	Multiple windows & frames, WebDriver Waits	Switching between windows	In-Person Trainer - ILT
		Switching to frames & iFrames	
		Implicit waits, Explicit waits, Fluent waits	
		Checkbox Handling and Form Submission	In-Person Trainer - ILT
		Locating Web Element Attributes using CSS Selector	
		Locating Web Element Attributes using Absolute XPath Locator	
Chapter 6	Selenium Alerts & Regular Expression	HandsON : TekStac Lab	DLE & BU Connects
		Identifying Ancestors of Web Elements	In-Person Trainer - ILT
		Working with alerts	
		Working with Java script executor	
		Working with Regular expression	
		Working with Drag and Drop and Slider	
Chapter 7	Advanced WebDriver techniques	HandsON : TekStac Lab	DLE & BU Connects
		Selenium - ScreenShot	In-Person Trainer - ILT
		Handling JavaScript/window Alerts	
		Headless browsers and drivers in Selenium	
		Handling Dynamic Elements in Selenium	
		HandsON : TekStac Lab	DLE & BU Connects
		1) get methods , 2) conditional methods, 3) browser methods, 4) navigational methods, 5) wait methods	In-Person Trainer - ILT
		How to Find All/Broken links using Selenium Webdriver	
		HandsON : TekStac Lab	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

3. Refer to and learn the same topics covered by the trainer on Udemy.
4. Practice the Tekstac Platform Hands-on.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- CSS Selector
- AbsoluteXpathLocator
- xPathAncestor
- Working With Alerts
- Work with Javascript Executor
- Handling RegEx Selenium
- Drag and Drop with Slider and Actions and wait
- Selenium - ScreenShot
- Selenium - DatePicker(Actions Class)
- FindElements
- Handling ToolTip
- Working with different XPath conditions
- PurchaseApp iframe
- Multiple Window
- Selenium - Implicit Wait
- Selenium - Fluent Wait

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Enhanced Automation Test Scripts

- Trainer-Guided Integrated Project: Comprehensive Error Handling and Reporting
- Trainer-Guided Integrated Project: Utility Class for Advanced Functions
- Trainer-Guided Integrated Project: Code Review & Refactoring Artifacts

QEA Role specific Automation Technical Handson

Mandatory Hands-on

- Chapter 6_Handson
- Chapter 7_Handson
- Chapter 8_Handson
- Chapter 9_Handson
- Chapter 10_Handson
- Chapter 11_Handson
- Desire_Capabilities
- Shadow_DOM_Scenario1
- Shadow_DOM_Scenario2
- Relative Locators and Chained Locators in Selenium_Scenario1
- Relative Locators and Chained Locators in Selenium_Scenario2
- Relative Locators and Chained Locators in Selenium_Scenario3
- Relative Locators and Chained Locators in Selenium_Scenario4

Once completing Udemy course, Learnings, all Handson & Practice assessments related to Selenium-Web Driver skill appear for the Code Challenge **Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge (Day 15)

Assess-Type-1:Code Challenge - Selenium Automation Techniques, Dynamic XPath

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

Selenium Webdriver With POM and ApachePOI

Learn the basics of WebDriver Basics



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 4 in this Udemy course and complete the corresponding Subtopics

Section 4: Apache POI - Data Driven Testing using MS Excel

Selenium - Course - TOC			
Chapter	Chapter Name	Topic	Training Mode
Chapter 8	Calendar using action & Apache POI	Selenium - DatePicker(Actions Class)	In-Person Trainer - ILT
		Apache POI - Read Operation : Excel Read Operation	
		Apache POI - Write Operation: Excel Write Operation	
		Applying POI POM- Flight reservation	
		HandsON : TekStac Lab	DLE & BU Connects

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Excel Read Operation
- Excel Write Operation
- Applying POI - Flight reservation
- POM with Page Factory Model

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Refactored Automation Framework (Page Object Model Implementation)
- Trainer-Guided Integrated Project: Data Management Strategy & Implementation (Apache POI)
- Trainer-Guided Integrated Project: Execution Confirmation and Custom Logging Artifacts
- Trainer-Guided Integrated Project: Updated Agile Artifacts & Traceability (Living Documentation)

QEA Role specific Automation Technical Handson

Mandatory Hands-on

- Chapter 12_Handson
- Mouse_Keyboard_Actions_Cookie_Scenario1
- Mouse_Keyboard_Actions_Cookie_Scenario2
- Mouse_Keyboard_Actions_Cookie_Scenario3
- Mouse_Keyboard_Actions_Cookie_Scenario4
- Mouse_Keyboard_Actions_Cookie_Scenario5

- Mouse_Keyboard_Actions_Cookie_Scenario6
- Mouse_Keyboard_Actions_Cookie_Scenario7

Stage : GenAI Code Assist (GitHub Copilot)

Learning and practicing the below provided GenAI topics will equip you with essential, modern skills in QA Automation, preparing you for an evolving tech landscape

- **GenAI Foundation**

This section provides the foundational knowledge of how GenAI works and the critical importance of using it ethically and responsibly.

- Fundamentals of Generative AI [101-Basics]
- Generative AI Quick Assessment for Elearning Quiz [101-Basics]
- Introduction to Responsible AI
- Responsible AI-Deep Dive [101- BASICS]

- **Prompt Engineering:** The ability to communicate effectively with AI models is a core competency for maximizing their utility. This skill directly translates to generating higher quality code suggestions, more relevant test cases, and accurate documentation, saving significant time and effort.

- Prompt Engineering Foundation - PY0009562

- **GitHub Copilot:** This section focuses on integrating AI assistance directly into the daily workflow. Learning to use GitHub Copilot effectively helps freshers write code faster, reduce bugs through AI-driven suggestions, and understand best practices through examples provided by the AI.

Introduction to GitHub Copilot

[GitHub Copilot for Beginners: AI Coding Crash Course](#)



Go through entire course. Implement the examples along with the author.

[GitHub Copilot - The Complete Guide – 2025](#)



Go through entire course. Implement the examples along with the author.

- **Usecase and Handbook & Comprehensive GenAI Hands-On for Automation Developers:** These exercises provide practical experience in optimizing AI outputs, managing token limitations, and generating functional,

high-quality automation code, making the learning directly applicable to their job roles.

Code Challenge (Day 20)

Once completing Udemy course, Learnings, all Handson & Practice assessments related to Selenium-Web Driver skill appear for the Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Assess-Type-1: Code Challenge - Selenium Webdriver With POM and ApachePOI

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

Day 21 to 25

Automation Testing - Selenium with TestNG



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer sections in this Udemy course and complete the corresponding Subtopics

Section 5: TestNG Framework for Selenium

Section 6: Page Object Model

Section 7: Hybrid Driven Framework Development with Project (eCommerce Application)

Selenium - Course - TOC			
Chapter	Chapter Name	Topic	Training Mode
Chapter 9	TestNG Concepts & TestNG Class & Data driven	TestNG Assertions	In-Person Trainer - ILT
		Introduction to TestNG	
		Creating TestNG test cases	
		TestNG annotations and assertions	
		Data-Driven Testing in TestNG	
		Running Tests in Parallel using TestNG	
		HandsON : TekStac Lab	DLE & BU Connects
		Multi-browser testing using TestNG*	In-Person Trainer - ILT
		Rerun failed tests*	
		Set Test priority in TestNG	
		Dependency in TestNG	
		Soft /Hard Assertion in TestNG	
		TestNG Report Generation in Selenium WebDriver*	
		Customize, PDF & Email TestNG Reports in Selenium WebDriver*	
		HandsON : TekStac Lab	DLE & BU Connects
		TestNG with two test classes	In-Person Trainer - ILT
		Data driven from XML + POM : Contact Book- POM structure	
		HandsON : TekStac Lab	DLE & BU Connects

Learn the basics of Selenium with Datasource XML Parsing (With XML & JSON)



[Learn API Technical Writing: JSON and XML for Writers](#)

Refer sections 1, 2, 3 and 4 in this Udemmy course and complete the corresponding learnings.

Section 1: Introduction

Section 2: JSON

Section 3: XML

Section 4: Final Words

Rehearse the below Lend a hand enablement given in platform

- XMLParser_Enablement

Followed by refer “Lend-a-Hand” code template with solution

- XML Parsing

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- TestNG Assertions
- TestNG with two test classes
- Contact Book- POM structure
- TestNG Annotation With Priority
- TestNG Dependency
- Cargo Shipping
- Shopify Registration
- XML Parsing 1
- XML Parsing 2
- XML Parsing 3
- JSON Parsing 1
- JSON Parsing 2
- JSON Parsing 3

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Integrated & Structured Test Suite using TestNG Framework
 - ✓ You should have automated more than one sprints
 - ✓ Ensure you have in-sprint automation concepts introduced as part of project
 - ✓ Ensure you have created Regression suite and smoke suite automation
 - ✓ Ensure your regression suite consists of minimum 25 test cases and test scripts
 - ✓ Ensure your smoke suite covers end to end business functionality

- Trainer-Guided Integrated Project: Robust Test Data Management via TestNG DataProviders
- Trainer-Guided Integrated Project: Comprehensive Execution Control via testng.xml
- Trainer-Guided Integrated Project: Advanced Reporting with Extent Reports & Listeners
- Trainer-Guided Integrated Project: Implementation of Assertions (Hard and Soft)

QEA Role specific Automation Technical Handson

Mandatory Hands-on

- Chapter 13

Once completing Udemy course, Learnings, all Handson & Practice assessments related to Selenium-Web Driver skill appear for the Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge (Day 25)

Assess-Type-1: Code Challenge - Selenium with TestNG

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

Day 26 to 30

Automation Framework

Chapter	Chapter Name	Topic
Chapter 14	Test Automation Frameworks	Introduction to Automation Frameworks
		Data-Driven Testing in Selenium
		Keyword-Driven Testing Framework in Selenium
		Page Factory
		Building a Modular Testing Framework in Selenium
		Hybrid Framework
	Design Patterns (POM)	Understanding Page Object Model (POM) in Selenium
		Creating Page Classes in Selenium
		Implementing Page Factory in Selenium
		Design Patterns in Automation using Selenium (Singleton, Factory etc.)
Chapter 15	Test Data Handling	Reading and Writing Data to Files (e.g., Excel, CSV , XML, JSON) using Java
		Test Data Parameterization in Selenium
Chapter 16	Test Report and Test Log	Generating Detailed Test Reports (Extent Reports, Allure Reports)
		Screenshot Capture Strategies in Selenium
		Custom Logging with Log4j in Selenium

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 7 in this Udemy course and complete the corresponding Subtopics

Section 7: Hybrid Driven Framework Development with Project (eCommerce Application)

Mandatory Hands-on

- Trainer-Guided Integrated Project: The "Sprint Automation" Hybrid Framework Codebase
- Trainer-Guided Integrated Project: Complete Automated Regression/Sprint Suite
- Trainer-Guided Integrated Project: Enterprise-Level Reporting Suite (Extent & Allure)
- Trainer-Guided Integrated Project: CI/CD Ready Execution Artifacts
- Trainer-Guided Integrated Project: Finalized Agile Documentation & Handover Package

QEA Role specific Automation Technical Handson

Mandatory Hands-on

- Chapter 14 & 15_Handson
- Chapter 16_Handson

Devops with BDD and Docker



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer sections in this Udemy course and complete the corresponding Subtopics

Section 8: CI - Maven, Git, Github & Jenkins

Selenium - Course - TOC			
Chapter	Chapter Name	Topic	Training Mode
Chapter 13	Selenium Grid & Distributed Execution	Introduction to Selenium Grid	In-Person Trainer - ILT
		Setting up Selenium Grid	
		Running tests on Selenium Grid	
		Achieving Parallel Execution in the Cloud	
		Setting up selenium grid in a docker/K8S container	DLE & BU Connects
		HandsON : TekStac Lab	
		Running Tests on Cloud-Based Selenium Grids (e.g., Sauce Labs, Browser Stack)	In-Person Trainer - ILT
		Achieving Parallel Execution in the Cloud	DLE & BU Connects
		HandsON : TekStac Lab	
Chapter 14	Maven & Git Version Control	Introduction to Maven	In-Person Trainer - ILT
		Creating and Managing Maven Projects	
		Maven Build Lifecycle (e.g., clean, compile, test, package)	
		Introduction to Version Control	
		Git Fundamentals (Commits, Branches, Merges)	
		Git Workflow (Pull Requests)	
		Managing Code Repositories on GitHub or GitLab	DLE & BU Connects
		HandsON : TekStac Lab	
Chapter 15	Jenkins CI Systems	Introduction to CI/CD and DevOps	In-Person Trainer - ILT
		Jenkins Overview and Jobs Configuration	
		Integrating Selenium Tests into Jenkins Pipelines	
		Groovy Script Introduction and Syntax	
		HandsON : TekStac Lab	DLE & BU Connects

Mandatory Hands-on

- Setup Grid and parallel execution
- Git Config
- Clone Repo
- Add, Commit And Push
- Pull And Merge
- Merge - Resolve Conflict
- Git Tags

Day 31 to 35

Devops with BDD and Docker

Selenium - Course - TOC			
Chapter	Chapter Name	Topic	Training Mode
Chapter 13	Selenium Grid & Distributed Execution	Introduction to Selenium Grid	In-Person Trainer - ILT
		Setting up Selenium Grid	
		Running tests on Selenium Grid	
		Achieving Parallel Execution in the Cloud	
		Setting up selenium grid in a docker/K8S container	DLE & BU Connects
		HandsON : TekStac Lab	
		Running Tests on Cloud-Based Selenium Grids (e.g., Sauce Labs, Browser Stack)	In-Person Trainer - ILT
		Achieving Parallel Execution in the Cloud	DLE & BU Connects
		HandsON : TekStac Lab	
Chapter 14	Maven & Git Version Control	Introduction to Maven	In-Person Trainer - ILT
		Creating and Managing Maven Projects	
		Maven Build Lifecycle (e.g., clean, compile, test, package)	
		Introduction to Version Control	
		Git Fundamentals (Commits, Branches, Merges)	
		Git Workflow (Pull Requests)	
		Managing Code Repositories on GitHub or GitLab	DLE & BU Connects
		HandsON : TekStac Lab	
Chapter 15	Jenkins CI Systems	Introduction to CI/CD and DevOps	In-Person Trainer - ILT
		Jenkins Overview and Jobs Configuration	
		Integrating Selenium Tests into Jenkins Pipelines	
		Groovy Script Introduction and Syntax	
		HandsON : TekStac Lab	DLE & BU Connects

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer sections in this Udemy course and complete the corresponding Subtopics

Section 10: Bonus topics

Section 11: SelectorsHub

Section 12: Wind up Session- Learning , Practice Plan & Interview Preparation Tips

Section 13: Docker - Bonus topics

Mandatory Hands-on

- Using parameters in Jenkins Pipeline
- Compile and run Java Application using Jenkins
- Deploy a webserver container using Jenkins
- Build Web Application
- Compiling Executing Java using POM
- Junit With Maven
- Maven Directory Structure
- Maven Shade Plugin
- Maven App to access External Service
- Capital Service
- Address Book
- Customer Registration
- Simple cucumber project with phptravels
- Simple cucumber project using datatable
- Simple cucumber project using scenario outline
- Java Service on Docker
- Docker - Running a Java Application
- Deploy Nginx and MySQL using docker-compose
- Docker - Running Static Sites in an Apache Server
- Deploy Apache php and MySQL using docker-compose

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

- Trainer-Guided Integrated Project: Scalable Execution Infrastructure (Selenium Grid & Docker)
- Trainer-Guided Integrated Project: Behavior-Driven Development (BDD) Framework Integration
- Trainer-Guided Integrated Project: Continuous Integration & Delivery (CI/CD) Pipeline Integration

QEA Role specific Automation Technical Handson

Mandatory Hands-on

- Chapter 17_Handson
- Chapter 18_Handson
- Chapter 19_Handson
- Chapter 21_Handson
- Chapter 22_Handson
- Chapter 23_Handson
- Chapter 24_Handson

Day 36 to 40

API Automation

Continuous Learning: ILT Technical Enablement

Refer Section 1,2,3 in the below Udemy course to understand the concepts

Note: Section : 4 is optional



[Learn API Testing using Postman, SoapUI & RestAssured](#)

Mandatory Hands-on

- User Service 1 - Test Using Postman
- User Service 2 - Test Using Postman
- User Service 3 - Test Using Postman
- Cinema Hall Service 1 - Test Using Postman
- Find Winner SOAP Webservice - Test Using Postman

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

- Trainer-Guided Integrated Project: The "API Smoke Suite" Postman Collection
- Trainer-Guided Integrated Project: The "Rest Assured" Automation Module (Framework Integration)
- Trainer-Guided Integrated Project: Comprehensive Automation Strategy & Traceability
- Trainer-Guided Integrated Project: Consolidated Execution Reports

QEA Role specific Automation Technical Handson

Mandatory Hands-on

- Verify the Success Status code 200 & 404 for the Bitcoin rate against USD and Non-USD currencies.
- Print Bitcoin rate against USD, GBP & EuR currency
- Download Postman
- Create workspace for API Request
- Create the Collections for API Request
- Sending the first API Request
- Get the response of GET API Request
- Get the response of POST API Request
- Get the response of PUT API Request
- Get the response of PATCH and DELETE API Request
- Create an environment in Postman
- Set the environment variables in Postman
- Get the values of the environment variables in Postman
- Get the ID of the given API in Postman
- Data Parameterization using CSV in Postman
- Test API Using Swagger UI
- Verify the response of the GET API
- Validate the 200 status code in Rest Assured
- Validate the 301 status code in Rest Assured
- Validate the 404 status code in Rest Assured
- Validate the 500 status code in Rest Assured
- Get the API response in JSON in Rest Assured
- Get the API response in XML in Rest Assured
- Get the API response with single query parameter in Rest Assured
- Get the API response with double query parameter in Rest Assured
- Get the API response with path parameter in Rest Assured
- Get the API response with base URI in Rest Assured
- Create workspace for API Request
- Create the Collections for API Request
- Sending the first API Request
- Get the response of GET API Request
- Get the response of POST API Request
- Get the response of GET API Request
- Get the response of PUT API Request
- Get the response of PATCH API Request
- Get the response of DELETE API Request
- Create an environment in Postman
- Set the environment variables in Postman
- Get the values of the environment variables in Postman
- Get the ID of the given API in Postman
- Data Parameterization using CSV in Postman
- Verify the Success Status code 200 for Add product details request
- Print the success response for Add product details request
- Update price of existing product details and verify the success status code 200
- Update price of existing product details and print the updated price from response

- Update only name of the existing product and verify the success code 200 displayed
- Delete the product which is not available in the server and verify the status code 404 and 200

Day 41 to 45

API Automation

Continuous Learning: ILT Technical Enablement

Refer Section 1,2,3 in the below Udemy course to understand the concepts

Note: Section : 4 is optional



[Learn API Testing using Postman, SoapUI & RestAssured](#)

QEA Role specific Automation Technical Handson

Mandatory Hands-on

- Verify the response status code with status code method in Rest Assured
- Verify the response status code with assert That in Rest Assured
- Verify the response status code with assertion with status Line method in Rest Assured
- Verify the response status code with assertion with Hamcrest matchers method in Rest Assured
- Handle the response using Getting a single header in Rest Assured
- Handle the response using Getting all headers in Rest Assured
- Validating a header in Rest Assured
- Validating multiple headers in Rest Assured
- Validating JSON schema using matchesJsonSchemaIn Classpath in Rest Assured
- Field-by-Field Validation in Rest Assured
- Structural Validation in Rest Assured
- Validate the inline schema in rest assured
- Validating response using Equality Check in Rest Assured
- Validating response using List Size Check in Rest Assured
- Validating response using Existence Check in Rest Assured
- Validating response using Multiple Assertions in Rest Assured
- Validating response using Logical Operations in Rest Assured
- Validating response using Custom Parsers in Rest Assured
- Validating response for a List of Values in Rest Assured
- Validate cookies in Rest Assured
- Get the response of an API using single header in Rest Assured
- Get the response of an API using multiple headers in Rest Assured
- Get the response of an API by setting headers from a map in Rest Assured

- Utilize the redirect false command in Rest Assured
- Limit the number of redirects in Rest Assured
- Follow redirects but not use relaxed HTTPS validation on the redirected requests in Rest Assured
- Ensure JSON schema data types are matching for the jsonplaceholder API request.
- Ensure JSON schema data types are not matched for the jsonplaceholder API request.
- Print number of universities in United States
- Print number of universities in India
- Print number of universities in Canada
- Print the university names in United States
- Print the university names in India
- Print the university names in Canada
- Store Uganda university details response into a json file
- Store Spain university details response into a text file
- Print second available university name in France
- Verify the United States Population is "318558162" in the year of 2016
- Verify the age is "34" for the name="meelad"

NOTE: On Day 45, Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to Selenium skill appear for the MOCK ICT Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Integrated Capability Test (ICT) - SELENIUM MOCK

Assessment Duration: 4 hrs.

Assessment Attempts: 1

No. of Questions appear in each attempt : 1

Day 46 to 47

GenAI Code Assist (GitHub Copilot)

Comprehensive GenAI Hands-On for Automation Developers

- Trainer-Guided Integrated Project: Chat Endpoint Basics
- Trainer-Guided Integrated Project: Understanding Tokenization & Context Limits (Estimate & Optimize Tokens)
- Trainer-Guided Integrated Project: Prompt Optimization via /ai/chat/with-usage
- Trainer-Guided Integrated Project: Generate Edge Cases
- Trainer-Guided Integrated Project: From User Story to Feature File (AI-Driven BDD Generation)
- Trainer-Guided Integrated Project: Generate Locators for SauceDemo Login (Dynamic Locator Generation)
- Trainer-Guided Integrated Project: AI-Augmented Framework
- Trainer-Guided Integrated Project: Prompt Engineering Game

NOTE: On Day 47, Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to Selenium skill appear for the ICT Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

SELENIUM Assess-Type-2: Integrated Capability Test (ICT)

Assessment Duration: 4 hrs.

Assessment Attempts: 1

No. of Questions appear in each attempt : 1

Day 48 to 50

Evaluation:

Final Integrated Project Evaluation + Final Technical Evaluation

The overall Final Technical evaluation will cover topics specified in the curriculum and will be conducted through a video interview on the GenC Learn Platform.

Trainees are given two attempts to pass the final evaluation. If a Trainee receives a RAG status of “RED” or “AMBER” status on their first attempt, they have one opportunity to improve and achieve a “GREEN” status on their second attempt.

What is Final Evaluation?

The Final Evaluation will be conducted to certify whether a GenC is eligible to enter into the BU or not. The skill of a GenC will be gauged on the application development and overall technical knowhow towards the end of GenC Training.

Tech SME from BU will be conducting the final tech evaluation. As a fallback, the project mentor can also steer this activity.

The final evaluation will be conducted as two phases. They are the following

- 1. Final Technical Evaluation**
- 2. Final Integrated Project Evaluation**

The mode of these evaluations will be any one of the following:

- F2F(face to face)
- Video Based

1. Final Technical Evaluation (FTE)



The BU Mentor will interview the GenC on various skills achieved throughout the training program and put a score which will be considered for the final PHS of the GenC.

2. Final Project Evaluation (FPE)

In this evaluation, the BU Mentor will be verifying the skills of a GenC on a project perspective. End of this evaluation, the BU Mentor will score the GenC's work based on various evaluation criterions.

FAQs

1. What is the HiPo GenC Accelerated Learning Program?

The HiPo (High Potential) GenC Accelerated Learning Program is a fast-track learning initiative for GenCs who demonstrate strong foundational and behavioral skills. It enables early graduation and deployment into business projects.

2. How do I get selected for the HiPo batch?

Selection is based on:

- Performance in the Boot Camp & Behavioural Assessment
- Results from the Foundational Technical Assessment
- Evaluation by Subject Matter Experts (SMEs) through 1:1 interview

3. What happens during Boot Camp?

The Boot Camp includes:

- Organizational learning sessions
- Behavioral workshops by certified trainers
- Leader connects sessions
- Stakeholder introductions
- Training program orientation
- Policy awareness sessions

4. What is the Foundational Technical Assessment?

It's a checkpoint to validate your readiness in:

Skill	Topics
Core Java	Java Programming fundamentals, OOPs, collections, exception handling, streams,
	functional programming

Data Structure Algorithms	Data Structures: Arrays, Array Lists, Linked Lists, Doubly Linked Lists,
	Stacks & Queues, Binary Trees, Hash Tables, Heaps, Graphs
	Algorithms: Dynamic Programming, Sorting, Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort
	Searching: Breadth First Search, Depth First Search
SQL	SQL Fundamentals, Data modeling, queries, joins, and constraints
HTML5, CSS3, JavaScript	Web structure, styling, and interactivity

***** Clearing this assessment is mandatory to enter the HiPo track*****

5. What will I learn in the HiPo track?

You'll learn:

- GenAI for QA Engineers and Hackathon project Automation
- Advanced Selenium Automation techniques
- GitHub Integration, TestNg Listeners and Parallel Execution
- Apache POI and BDD Cucumber Integration
- Devops Integration, Git Merge Conflicts and Jenkins Jobs
- Design Frameworks from scratch
- Regression suite Test Automation Reporting and Customization
- API Automation and RestAssured Automation

6. What is Gen AI Learning?

It's a mandatory module that introduces you to generative AI tools and practices, helping you become a future-ready developer.

7. Will I interact with business leaders or mentors?

Yes! You'll participate in Business Mentor Connect sessions to gain real-world insights and understand customer expectations.

8. What happens after I complete the program?

If you successfully complete all evaluations, you'll be:

- Graduated early
- Deployed to niche customer engagements
- Recognized as a HiPo GenC with accelerated career opportunities